OpenShift Agent-Install Ecosystem

The components that together provide OpenShift agent-based installs on the home networks (g8 multi-node on Proxmox, g10 single-node on bare metal). Each is its own repo; this doc is the map.

There are **two working install flows** sharing the same core services:

Flow	Project	Target	Registry
Multi-node (3 control + 3 worker VMs)	`agentinstall`	Proxmox `pve.g8.lo` , VMs 700–706	`fastregistry.g8.lo:5000`
Single-node (SNO) bare metal	`agent-sno-baremetal`	`server1.g10.lo` via PXE	`fastregistry.g10.lo:5000`

Both flows: mirror a release into FastRegistry → generate the agent ISO server-side on FastRegistry → boot the target → watch progress in agent-monitor.

Components

agentinstall — multi-node install workflow (this repo)

Repo: `git@github.com:glennswest/agentinstall.git`

Shell-script-driven agent-based install of a 6-node cluster (`g8` / `g8.lo` , rendezvous `192.168.8.201`) as Proxmox VMs on `pve.g8.lo` (control 701–703, worker 704–706, `production-lvm` storage). All settings live in `config.sh` . Cached `openshift-install` Mac binaries for 4.18–4.21 are kept in `bin/` .

Nodes PXE-boot the FastRegistry-generated agent ISO via the PXE Manager — no ISO is attached to the VMs. See `README.md` for the full step-by-step flow, configuration reference, and troubleshooting.

Usage:

```
./generate-secrets.sh      # first time only — pull secret + SSH key
./mirror.sh 4.18           # mirror release via FastRegistry API (resolves latest 4.18.x)
./create-vms.sh            # first time only — create VMs 700-706
./install.sh 4.18          # generate configs, ISO via FastRegistry, boot nodes, monitor
```

Supporting scripts: `verify.sh` (post-install checks), `watch-install.sh` , `approvecsr.sh` , `poweron-all.sh` / `poweroff-all.sh` , `delete-vms.sh` , `gatherdebug.sh` , `wipe_registry.sh` , `remirror.sh` .

agent-sno-baremetal — single-node bare-metal install workflow

Repo: `git@github.com:glennswest/agent-sno-baremetal.git`

Fully API-driven SNO install onto a bare-metal server (`server1.g10.lo` , cluster `s1.lo`). One script orchestrates the whole g10 service stack: FastRegistry (release + ISO), PXE Manager (`pxe.g10.lo:8080`) for network boot, netman (`network.g10.lo`) for DNS/DHCP provisioning, and agent-monitor for progress. Configuration in `config.sh` , install-config/agent-config templates in `templates/` .

Usage:

```
# edit config.sh (server, cluster name, versions), then:
./install-sno.sh
```

agent-monitor — the install console

Repo: `git@github.com:glennswest/agent-monitor.git` (v1.0.1)

Go + HTMX web UI ("Agent Install Monitor") that watches installs live. It polls the assisted-installer REST API the agent installer exposes on the rendezvous node (`http://<rendezvous-ip>:8090/api/assisted-install` , JWT via `Watcher-Authorization` header) through a connect → monitor → terminal lifecycle. Multi-cluster tabs; per-cluster overview, hosts, validations, and events views. Deployed as an ARM64 container at `http://agentmonitor.g10.lo` (port 80) on MikroTik rose via mkpod; images built by GHCR CI.

Usage:

```
make run                    # local dev on :8080
make redeploy               # build + deploy to rose via mkpod (deploy.sh wraps
deploy_agent_monitor.py)
```

In the UI, "Add cluster" takes a name, the rendezvous IP, and the token (printed by `openshift-install agent create image` / found in the install workdir auth assets).

fastregistry — registry, release mirror, and agent-ISO factory

Repo: `git@github.com:glennswest/fastregistry.git` (v0.6.x)

Go container registry with a release manager: it discovers and clones OCP release payloads, extracts client artifacts (`oc` , `openshift-install` , `coreos.iso`), and — the key piece for agent installs — **generates agent ISOs server-side** so clients never download the ~1 GB base ISO. Send it ~10 KB of configs; it builds the ignition (`internal/releases/ignition.go`) and embeds it with `coreos-installer` (`internal/releases/iso.go`). Instances: `fastregistry.g8.lo:5000` and `fastregistry.g10.lo:5000` (systemd service; HTTP, no auth).

Usage (ISO generation API):

```
curl -X POST "http://fastregistry.g8.lo:5000/admin/releases/4.21.0-x86_64/iso" \
  -H "Content-Type: application/json" \
  -d "{
    \"install_config\": $(cat install-config.yaml | jq -Rs),
    \"agent_config\": $(cat agent-config.yaml | jq -Rs)
```

```
  }"
# → {"full_url": "http://.../files/installs/<uuid>/agent.iso"}  — PXE-sanboot ready
```

Mirroring and extraction are driven through its admin API (wrapped by `agentinstall/mirror.sh`); a web UI (releases, repositories, sync, download stats) runs on the same port.

pxemanager — PXE boot service (SNO flow)

Repo: `github.com/glennswest/pxemanager`

Lightweight PXE boot manager with an HTMX web UI, IPMI integration, and a Redfish API (Bare Metal Operator compatible). Runs at `pxe.g10.lo:8080`; `agent-sno-baremetal` points it at the FastRegistry-generated agent ISO to network-boot the target server.

quick-quay — legacy registry path (superseded)

Repo: `git@github.com:glennswest/quick-quay.git`

Native (non-container) Quay install on Fedora at the old `registry.gw.lo`, with release-mirroring scripts. This was the registry behind agentinstall before the FastRegistry migration (`Switch from Quay to FastRegistry` commit). Kept for reference; new work uses fastregistry.

Related / adjacent

- **rspaced** (`github.com/glennswest/rspace` workspace) — experimental bootc-based boot agent intended to replace RHCOS in the agent-installer flow (live boot, pull from rspacefs, pivot without reboot). Research direction, not part of the working install path.
- **netman** — DNS/DHCP provisioning API used by the SNO flow.
- **microdns** — per-network DNS/DHCP servers backing all of the above (cluster records: `api.<cluster>.<domain>`, `*.apps.<cluster>.<domain>`).

End-to-end: multi-node install on g8

```
cd ~/projects/agentinstall
./mirror.sh 4.21                    # once per release
./install.sh 4.21
# watch at http://agentmonitor.g10.lo (rendezvous 192.168.8.201)
./verify.sh
```

End-to-end: SNO install on g10

```
cd ~/projects/agent-sno-baremetal
./install-sno.sh                    # mirrors, ISO, PXE, DNS/DHCP, monitor — all API-
driven
```